Identifiability-Aware Source Apportionment in City-Scale Advection-Diffusion Systems

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Abstract

Source apportionment from sparse urban air-quality sensors is an inverse problem whose resolution is limited by sensor placement, wind-driven transport, background variation, and noise. Known or proxy emission inventories make attribution scientifically meaningful by restricting the unknown source field to a finite set of candidate source groups, but they do not guarantee that those groups are distinguishable from the available observations. We represent time-varying source activity with a low-dimensional nonnegative temporal basis and formulate inventory-based source apportionment as a wind-conditioned lagged inverse problem in which each source–basis coefficient produces a sensor-time fingerprint. After projecting out a separate low-dimensional background space, the relevant object is the projected lagged source–basis response matrix ($\widetilde{H}_\Phi$). We show that exact identifiability at the chosen temporal-basis resolution requires full column rank of $\widetilde{H}_\Phi$, while noise-robust attribution is controlled by its singular values, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance. Building on this analysis, we propose an identifiability-aware apportionment (IASA) framework that estimates nonnegative source–basis coefficients, reconstructs source activity trajectories, and reports the finest source grouping supported by the sensing system, including uncertainty intervals and merge recommendations for indistinguishable source groups. Controlled city-scale advection–diffusion experiments evaluate whether attribution error, ambiguity, and the benefit of source merging are predicted by the geometry of $\widetilde{H}_\Phi$. The resulting framework treats sparse-sensor source apportionment not as a problem of producing the finest possible attribution vector, but as one of reporting the finest defensible attribution resolution.

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Preprint. June 20, 2026.

1. Introduction

Urban air pollution is driven by local emissions, meteorological transport, regional background pollution, and sensor noise. For policy, the central question is often not only *where* pollution is high, but *which* source groups contributed to the observed concentrations. Source apportionment therefore seeks to estimate the contributions of categories such as traffic, industry, brick kilns, road dust, and population-related activity from a combination of source inventories, transport models, and concentration measurements.

This task is especially difficult in sparse sensor networks. A model may fit sensor trajectories accurately while still failing to identify the underlying source contributions. The reason is geometric: after wind transport and sparse observation, two distinct source groups can induce nearly identical sensor-time signatures. A strong distant source and a weaker nearby source, for example, may become observationally equivalent once transport attenuation, finite travel time, background correction, and noise are taken into account. In such cases, fine-grained source attribution is not merely statistically uncertain; it is not identifiable from the available sensing system.

We study this issue in the inventory-based setting. Instead of attempting to recover an arbitrary unknown source field, we assume access to a finite set of known or proxy source maps. These inventories restrict attribution to a lower-dimensional source model and make the problem interpretable. To allow realistic time variation without estimating one free value at every hour, we represent each source activity as a nonnegative combination of a small number of temporal basis functions. The unknowns are therefore source–basis coefficients, from which source activity trajectories and contributions are reconstructed. However, the inventory and temporal-basis restrictions alone do not make the problem identifiable. Coefficient fingerprints can still be weakly visible, absorbed by background components, or mutually coherent after transport to the sensor network.

The key object in our analysis is a projected lagged source-response matrix. For a fixed wind sequence, transport operator, source inventory, sensor layout, and lag window, each source–basis pair induces a finite-horizon sensor-space fingerprint. Stacking these fingerprints gives a lagged source–basis response matrix H_{Φ}^{lag}. Real observations also contain background and temporal components, represented by a low-dimensional basis Q. After projecting out this background space, the inverse problem is governed by

$$\widetilde{H}_{\Phi} = P_Q^{\perp} H_{\Phi}^{\mathrm{lag}},$$

where $P_Q^{\perp}$ is the projection onto the orthogonal complement of the background basis. The rank, singular values, coefficient visibility, background absorption, pairwise coherence, and ray distance of $\widetilde{H}_{\Phi}$ determine which temporal source contributions can be separated and which should be reported only after merging or qualification. The familiar constant-activity model is recovered by using one constant basis function per source.

This perspective changes the role of source apportionment models. The goal is not to return the most detailed vector of source contributions permitted by the inventory. It is to report the finest source resolution that is defensible under the available sensors, wind regime, background correction, and noise level. An apportionment method should therefore output source estimates alongside identifiability diagnostics and ambiguity certificates.

We make the following contributions:

- We formulate sparse-sensor, inventory-based source apportionment as a lagged wind-conditioned inverse problem over nonnegative source–temporal- basis coefficients.
- We identify the projected lagged response matrix $\widetilde{H}_{\Phi} = P_Q^{\perp} H_{\Phi}^{\mathrm{lag}}$ as the central object governing coefficient separability after transport, sparse observation, lag, and background correction.
- We derive diagnostics for exact and noise-robust identifiability, including rank, smallest singular value, condition number, effective rank, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance.
- We propose an identifiability-aware apportionment procedure that fits nonnegative source–basis coefficients, reconstructs source activity trajectories, reports uncertainty, flags weakly visible coefficients, and merges source groups whose fingerprints are indistinguishable under the current sensing system.
- We design controlled city-scale advection–diffusion experiments to test whether source attribution error, source ambiguity, and the benefit of merging are predicted by the geometry of $\widetilde{H}_{\Phi}$ under varying noise levels, wind regimes, background bases, and sensor layouts.

The broader message is that sparse-sensor source apportionment is limited by the information content of the sensing system. Reliable attribution requires not only a transport model and an optimizer, but an explicit account of which source–basis fingerprints are visible and separable in the observed sensor space.

2. Related Work

Air-pollution source apportionment. Source apportionment has a long history in air-quality science, with receptor-oriented methods estimating source contributions from ambient chemical or physical measurements and source-oriented methods propagating emissions inventories through atmospheric transport models. Classical receptor frameworks include chemical mass balance and factor-analytic approaches such as positive matrix factorization (Watson et al., 2002; Paatero & Tapper, 1994; Hopke, 2016). Reviews and harmonization efforts have clarified best practices for receptor models, source-oriented models, and their combined use, while also documenting practical sensitivity to source profiles, factor interpretation, chemical resolution, and model comparability (Viana et al., 2008; Belis et al., 2013; 2019; Mircea et al., 2020; Hopke et al., 2020). This ambiguity is especially visible in settings such as India, where studies can produce divergent source-contribution estimates for the same city because of limited local profiles, collinear tracers, and methodological differences (Pant & Harrison, 2012); global reviews further show that policy-relevant source categories such as traffic, industry, dust, and domestic combustion are routinely reported from heterogeneous apportionment studies (Karagulian et al., 2015). Our work is complementary to this literature: rather than proposing another receptor model or emissions inventory, we analyze when known or proxy source maps are distinguishable from sparse concentration sensors after wind-conditioned transport, finite lags, background correction, and noise.

Data Assimilation and Inverse Problems Estimating latent system parameters from partial observations has been extensively studied in inverse problems and data assimilation (Tarantola, 2005; Engl et al., 1996). Classical approaches such as Kalman filtering, ensemble Kalman filters (Wang et al., 2023), and variational data assimilation methods (Carrassi et al., 2018) aim to infer hidden states and parameters from observations. These methods rely on known dynamical models and often require careful tuning of priors, covariance structures, and noise assumptions. However, these approaches typically assume that the underlying system is sufficiently observable, such that latent states and parameters can be inferred from available measurements. In sparse sensing regimes, this assumption breaks down, and multiple parameter configurations may remain consistent with the observed data, leading to fundamental nonidentifiability.

Physics-Informed Machine Learning Physics-informed neural networks (PINNs) and neural operators integrate governing equations into learning frameworks, enabling parameter estimation and system identification from data (Raissi et al., 2019; Li et al., 2024; 2020; Bhardwaj et al., 2025). These approaches enforce physical constraints through loss functions or architectural priors, allowing for flexible modeling of complex systems. Despite their success, when applied to inverse problems, these approaches implicitly assume that the inverse problem is well-posed, and that the available observations are sufficient to uniquely constrain the underlying parameters. Under sparse sensing, however, this assumption may fail, as distinct parameterizations can produce identical observational trajectories, limiting the effectiveness of purely optimization-based approaches.

Spatio-Temporal Models Graph neural networks (GNNs), transformers, and sequence models have been widely used to model spatio-temporal processes from sensor data (Li et al., 2017; Yu et al., 2017; Wu et al., 2019; Chen et al., 2023; Iyer et al., 2022). These models learn representations directly from observations without explicit reliance on governing equations. While these models effectively capture correlations in sensor observations, they operate purely in observation space and do not address whether the underlying physical systems are identifiable from such data. As a result, they cannot distinguish between observationally equivalent physical models.

Observability and System Identification The question of whether system parameters can be recovered from observations is closely related to classical notions of observability in control theory (Chen & Chen, 1984). For linear dynamical systems, observability characterizes whether the internal state can be uniquely determined from output trajectories. Extensions to nonlinear systems and PDEs have been studied in system identification and inverse problem literature (Ljung et al., 1987; Banks & Kunisch, 2012; Colton et al., 1990), often under assumptions of sufficient measurement richness. In contrast, we focus on the regime of sparse spatial sensing, where the observation operator induces a low-dimensional projection of the system, and analyze when such projections lead to indistinguishable dynamics across different physical models.

3. Problem Setup: Sparse Sensors, Wind, and Source Apportionment

We formalize source apportionment as a finite-dimensional inverse problem in which known source inventories are transported to a sparse sensor network under a wind-conditioned lagged operator. This section defines the observation model, the inventory parameterization, the background correction, and the projected response matrix used throughout the paper.

City-scale transport model. Let $u(\mathbf{x}, t) \in \mathbb{R}$ denote the pollutant concentration field on $\Omega \subset \mathbb{R}^d$. A generic continuous-time model is

$$\frac{\partial u}{\partial t} = \mathcal{F}_\eta(u, \mathbf{x}, t; w(t)) + s(\mathbf{x}, t) + b(\mathbf{x}, t), \tag{1}$$

where $w(t)$ is the meteorological state, s is local source emission, b is background or regional pollution, and η denotes transport and dispersion parameters. The exact dynamics may be approximated by a plume model, an advection–diffusion solver, or a learned surrogate constrained by transport structure.

Sparse observations. Discretize the concentration field on n grid cells, so that $\mathbf{u}_t \in \mathbb{R}^n$. Let $X_{\text{sens}} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be the fixed sensor locations with $m \ll n$. The observation operator $O : \mathbb{R}^n \to \mathbb{R}^m$ samples or interpolates the latent field at these locations:

$$\mathbf{y}_t = O\mathbf{u}_t + \boldsymbol{\epsilon}_t, \qquad \mathbf{y}_t \in \mathbb{R}^m. \tag{2}$$

Stacking over $t = 1, \dots, T$ gives

$$Y = [\mathbf{y}_1^\top, \dots, \mathbf{y}_T^\top]^\top \in \mathbb{R}^{mT}. \tag{3}$$

Because m is small relative to the field dimension, many latent fields and source configurations can be consistent with the same sensor trajectory.

Inventory-based source parameterization. We assume a set of K known or proxy source maps,

$$S = [\mathbf{s}_1 \ \mathbf{s}_2 \ \cdots \ \mathbf{s}_K] \in \mathbb{R}^{q \times K}, \tag{4}$$

where $\mathbf{s}_k \in \mathbb{R}^q$ is the spatial pattern of source group k. The source grid dimension q may differ from the concentration-grid dimension n. We represent time-varying source activity with a shared low-dimensional temporal basis

$$\Phi = [\boldsymbol{\phi}_1 \ \cdots \ \boldsymbol{\phi}_B] \in \mathbb{R}_+^{T \times B}, \qquad C = [c_{kb}] \in \mathbb{R}_+^{K \times B}. \tag{5}$$

The activity of source k at time t is

$$\theta_k(t) = \sum_{b=1}^{B} c_{kb} \phi_b(t), \qquad \theta_k(t) \geq 0. \tag{6}$$

The activity bases used for source emissions are nonnegative; signed temporal harmonics, when used, belong to the separate background basis Q. We vectorize C in source-major, basis-minor order as $\mathbf{c} = \text{vec}_{\text{src}}(C) \in \mathbb{R}_+^J$, where $J = KB$. The constant-window model is the special case $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$.

Lagged wind-conditioned response. Emissions released at time $t-\ell$ may affect sensors at time t only after being transported by the intervening wind field. Let

$$G^{\partial}_{t,t-\ell}(w_{t-\ell:t}) \in \mathbb{R}^{n\times q} \tag{7}$$

map emissions released on the source grid at time $t-\ell$ to the concentration grid at time t. The superscript ∂ denotes an open-boundary operator: mass transported outside Ω leaves the modeled domain rather than being reflected or wrapped around. For a maximum lag L, define

$$\mathcal{L}_t = \{0, 1, \ldots, \min(L, t-1)\}. \tag{8}$$

For source k and basis component b, the corresponding sensor response at time t is

$$\mathbf{h}^{\mathrm{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell) O G^{\partial}_{t,t-\ell}(w_{t-\ell:t})\mathbf{s}_k \in \mathbb{R}^m. \tag{9}$$

Stacking over time and arranging columns in source-major, basis-minor order gives

$$\begin{aligned} H^{\mathrm{lag}}_{\Phi} &= [\mathbf{h}^{\mathrm{lag}}_{11} \cdots \mathbf{h}^{\mathrm{lag}}_{1B} \cdots \mathbf{h}^{\mathrm{lag}}_{KB}] \in \mathbb{R}^{mT\times J}, \\ J &= KB, \end{aligned} \tag{10}$$

where each $\mathbf{h}^{\mathrm{lag}}_{kb} \in \mathbb{R}^{mT}$ is the finite-horizon source–basis, or coefficient, fingerprint. When $B = 1$, these columns reduce to the source fingerprints of the constant-activity model.

Background and temporal correction. Observed concentrations include components not explained by the local source inventory, including regional background, smooth temporal trends, and sensor-specific offsets. We represent these components with a low-dimensional basis

$$Q \in \mathbb{R}^{mT\times r}, \qquad \boldsymbol{\gamma} \in \mathbb{R}^r, \tag{11}$$

and write

$$Y = H^{\mathrm{lag}}_{\Phi}\mathbf{c} + Q\boldsymbol{\gamma} + E. \tag{12}$$

Let

$$P_Q^{\perp} = I - QQ^{\dagger} \tag{13}$$

be the projection onto the orthogonal complement of the background space, with $Q^{\dagger}$ denoting the Moore–Penrose pseudoinverse. The corrected inverse problem is

$$\widetilde{Y} = \widetilde{H}_{\Phi}\mathbf{c} + \widetilde{E}, \qquad \widetilde{Y} = P_Q^{\perp} Y, \qquad \widetilde{H}_{\Phi} = P_Q^{\perp} H^{\mathrm{lag}}_{\Phi}. \tag{14}$$

All identifiability diagnostics in this paper are computed from $\widetilde{H}_{\Phi}$, not from the raw inventory matrix or the unprojected response.

Projected source–basis fingerprints. Writing

$$H^{\mathrm{lag}}_{\Phi} = [\mathbf{h}^{\mathrm{lag}}_{11} \cdots \mathbf{h}^{\mathrm{lag}}_{KB}], \qquad \widetilde{H}_{\Phi} = [\widetilde{\mathbf{h}}_{11} \cdots \widetilde{\mathbf{h}}_{KB}], \tag{15}$$

we have $\widetilde{\mathbf{h}}_{kb} = P_Q^{\perp}\mathbf{h}^{\mathrm{lag}}_{kb}$. These projected fingerprints encode the observable effect of each temporal component of each source after wind transport, lag, sparse sensing, and background correction. Source apportionment at the chosen temporal-basis resolution is well posed only to the extent that these coefficient fingerprints are visible and separable.

4. Wind-Conditioned Response Construction and Estimation

Section 3 defines the projected inverse problem. This section describes how the response matrix is constructed in practice and how source activities are estimated once the response has been formed.

4.1. Coordinate-Aware Wind Field Estimation

Wind is usually observed at a small number of locations and must be estimated on the concentration grid. Let $X_{\mathrm{wind}} = \{\mathbf{z}_1, \ldots, \mathbf{z}_p\}$ be the wind-monitoring locations, and let $\mathbf{w}^{\mathrm{obs}}_t(\mathbf{z}_i) \in \mathbb{R}^2$ denote the observed horizontal wind vector at time t. We estimate a gridded wind field by

$$\widehat{\mathbf{w}}_{1:T} = f_{\phi}(\mathbf{w}^{\mathrm{obs}}_{1:T}, X_{\mathrm{wind}}, X_{\mathrm{grid}}). \tag{16}$$

The interpolator may be kriging, inverse-distance weighting, graph interpolation, or a local attention model over nearby wind observations. Since hourly wind observations already provide temporal structure, the interpolator uses a short temporal patch and explicit coordinates rather than an unconstrained long-horizon sequence model.

We use a residual parameterization around a physically reasonable baseline,

$$\widehat{\mathbf{w}}_t(\mathbf{x}) = \mathbf{w}^0_t(\mathbf{x}) + \Delta_{\phi}(\mathbf{x}, t), \tag{17}$$

where $\mathbf{w}^0_t$ is a baseline interpolation or reanalysis-guided wind field, and Δ_{ϕ} is a bounded correction. The wind model is trained on wind measurements before fitting pollution sensors:

$$\begin{aligned} \mathcal{L}_{\mathrm{wind}} = \sum_t \sum_{\mathbf{z}_i\in X_{\mathrm{wind}}} &\left\|\widehat{\mathbf{w}}_t(\mathbf{z}_i) - \mathbf{w}^{\mathrm{obs}}_t(\mathbf{z}_i)\right\|_2^2 \\ &+ \lambda_{\mathrm{sm}} R_{\mathrm{sm}}(\widehat{\mathbf{w}}) + \lambda_{\Delta}\|\Delta_{\phi}\|_2^2. \end{aligned} \tag{18}$$

A grid smoothness penalty,

$$R_{\mathrm{sm}}(\widehat{\mathbf{w}}) = \sum_t \sum_{(g,g')\in E_{\mathrm{grid}}} \|\widehat{\mathbf{w}}_t(\mathbf{x}_g) - \widehat{\mathbf{w}}_t(\mathbf{x}_{g'})\|_2^2, \tag{19}$$

discourages cell-to-cell oscillations while allowing spatial variation driven by measurements and coordinates. The residual penalty prevents wind from becoming a flexible latent variable chosen only to fit pollution sensors.

4.2. Open-Boundary Lagged Plume Response

The transport response $G^{\partial}_{t,\tau}$ maps emissions released at $\tau \leq t$ to the concentration field at time t. For any nonnegative emission vector $\mathbf{z} \in \mathbb{R}^q_+$, the open-boundary response obeys

$$G^{\partial}_{t,\tau}\mathbf{z} \geq 0, \qquad \mathbf{1}^\top G^{\partial}_{t,\tau}\mathbf{z} \leq \mathbf{1}^\top \mathbf{z}. \tag{20}$$

Thus the operator preserves nonnegativity and is mass non-increasing inside the modeled domain.

We instantiate this response with a differentiable puff approximation. For a unit release from source cell i at location $\mathbf{r}_i$ and release time τ, the puff center is initialized as $\mathbf{z}_i(\tau) = \mathbf{r}_i$ and advected by the interpolated wind field,

$$\frac{d\mathbf{z}_i(a)}{da} = \widehat{\mathbf{w}}_a(\mathbf{z}_i(a)), \qquad a \in [\tau, t]. \tag{21}$$

With substep size δa, this is discretized as

$$\mathbf{z}_i^{r+1} = \mathbf{z}_i^r + \delta a\, \widehat{\mathbf{w}}_{a_r}(\mathbf{z}_i^r). \tag{22}$$

If the puff exits Ω, its in-domain mass is removed. Otherwise its contribution is spread with an anisotropic Gaussian kernel

$$K_\psi(\mathbf{x}; \mathbf{z}_i, \Sigma_i) = \frac{\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{z}_i\|^2_{\Sigma_i^{-1}}\right)}{2\pi|\Sigma_i|^{1/2}}, \tag{23}$$

where ψ denotes dispersion parameters. A simple covariance model is

$$\Sigma_i(t,\tau) = R_i(t,\tau)\begin{bmatrix}\sigma^2_{\parallel}(t-\tau) & 0 \\ 0 & \sigma^2_{\perp}(t-\tau)\end{bmatrix} R_i(t,\tau)^\top, \tag{24}$$

with R_i aligned to the mean wind direction along the trajectory. The grid response to a unit release is

$$\left[G^{\partial}_{t,\tau}(\widehat{\mathbf{w}}_{\tau:t}; \psi)\mathbf{e}_i\right]_g = \chi_i(t,\tau) K_\psi(\mathbf{x}_g; \mathbf{z}_i(t), \Sigma_i(t,\tau))\Delta V_g, \tag{25}$$

where $\chi_i(t,\tau)$ indicates whether the puff remains active in the open domain and ΔV_g is the grid-cell area or volume. Kernel mass outside Ω is not renormalized.

4.3. Response Matrix and Source Activity Fit

Given the estimated wind field, source inventory, and temporal activity basis, the lagged source–basis response is constructed as in Section 3. Its (k, b)-th column is

$$\mathbf{h}^{\text{lag}}_{t,kb} = \sum_{\ell \in \mathcal{L}_t} \phi_b(t-\ell) O G^{\partial}_{t,t-\ell}(\widehat{\mathbf{w}}_{t-\ell:t}; \psi)\mathbf{s}_k, \tag{26}$$

and stacking over time and source-major, basis-minor columns yields $H^{\text{lag}}_{\Phi} \in \mathbb{R}^{mT \times J}$, $J = KB$. Background correction gives

$$\widetilde{Y} = P_Q^{\perp} Y, \qquad \widetilde{H}_\Phi = P_Q^{\perp} H^{\text{lag}}_{\Phi}. \tag{27}$$

Source–basis coefficients are then estimated by nonnegative regularized least squares,

$$\widehat{\mathbf{c}} = \arg\min_{\mathbf{c} \geq 0} \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda R(\mathbf{c}). \tag{28}$$

Reshaping $\widehat{\mathbf{c}}$ into $\widehat{C}$ reconstructs the source activity trajectory

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb}\phi_b(t). \tag{29}$$

The regularizer can be ridge stabilization,

$$R(\mathbf{c}) = \|\mathbf{c}\|_2^2, \tag{30}$$

or a weak prior around inventory-derived activity levels,

$$R(\mathbf{c}) = \|\mathbf{c} - \mathbf{c}_0\|_2^2. \tag{31}$$

Equivalently, one may fit background and source coefficients jointly:

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\gamma}}) = \arg\min_{\mathbf{c} \geq 0, \boldsymbol{\gamma}} \|Y - H^{\text{lag}}_{\Phi}\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda R(\mathbf{c}). \tag{32}$$

The projected formulation is used for identifiability because it exposes the part of each source–basis fingerprint that remains after background correction.

4.4. Constrained End-to-End Refinement

The most interpretable estimator fixes wind fields and transport parameters before fitting source activity. When wind interpolation or dispersion is uncertain, we optionally refine wind residuals and dispersion parameters under explicit constraints. Let ϕ_0 and ψ_0 denote the pretrained wind and initial dispersion parameters. The refinement objective is

$$\begin{aligned}\mathcal{L}_{\text{refine}} = &\|Y - H^{\text{lag}}_{\Phi}(\phi,\psi)\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda_\theta R(\mathbf{c}) \\ &+ \lambda_w \|\widehat{\mathbf{w}}^{\phi}_{1:T} - \widehat{\mathbf{w}}^{\phi_0}_{1:T}\|_2^2 + \lambda_\psi \|\psi - \psi_0\|_2^2 \\ &+ \lambda_{\text{sm}} R_{\text{sm}}(\widehat{\mathbf{w}}^{\phi}),\end{aligned} \tag{33}$$

subject to

$$\mathbf{c} \geq 0, \quad \psi \in \Psi_{\text{phys}}, \quad \|\widehat{\mathbf{w}}^{\phi}_{1:T} - \widehat{\mathbf{w}}^{\phi_0}_{1:T}\|_\infty \leq \epsilon_w. \tag{34}$$

This refinement is accepted only as a constrained local correction. Without these restrictions, wind and plume parameters could fit sparse pollution sensors while weakening the physical and source-level interpretation of $\widehat{\mathbf{c}}$ and the reconstructed activities.

5. Identifiability Theory for Source Apportionment

We now characterize when source activity at a chosen temporal-basis resolution is determined by the projected observations. Throughout this section,

$$\begin{aligned} \widetilde{Y} &= \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}, \\ \widetilde{H}_\Phi &= P_Q^\perp H_\Phi^{\mathrm{lag}} \in \mathbb{R}^{mT\times J}, \\ J &= KB. \end{aligned} \tag{35}$$

The analysis is conditional on the constructed wind field, transport operator, source inventories, temporal basis, lag window, and background basis. We use the flattened coefficient index $j = (k, b)$; "source fingerprint" without a basis qualifier refers only to the constant-activity case $B = 1$.

5.1. Exact Identifiability

Definition 5.1 (Exact identifiability)**.** The source–basis coefficient vector is identifiable from $\widetilde{Y}$ if

$$\widetilde{H}_\Phi \mathbf{c}_1 = \widetilde{H}_\Phi \mathbf{c}_2 \quad \Longrightarrow \quad \mathbf{c}_1 = \mathbf{c}_2. \tag{36}$$

This is identifiability of the reconstructed source activity trajectories at the selected temporal-basis resolution.

Proposition 5.2 (Identifiability of inventory-based apportionment)**.** *In the noiseless projected model $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c}$, $\mathbf{c} \in \mathbb{R}^J$ is identifiable for all coefficient vectors if and only if*

$$\operatorname{rank}(\widetilde{H}_\Phi) = J. \tag{37}$$

Proof. If $\operatorname{rank}(\widetilde{H}_\Phi) = J$, then $\ker(\widetilde{H}_\Phi) = \{0\}$, so $\widetilde{H}_\Phi(\mathbf{c}_1 - \mathbf{c}_2) = 0$ implies $\mathbf{c}_1 = \mathbf{c}_2$. Conversely, if $\operatorname{rank}(\widetilde{H}_\Phi) < J$, there exists a nonzero $\Delta\mathbf{c} \in \ker(\widetilde{H}_\Phi)$. Then $\widetilde{H}_\Phi \mathbf{c} = \widetilde{H}_\Phi(\mathbf{c} + \Delta\mathbf{c})$, so the coefficient vector is not uniquely determined. □

Nonnegativity can remove some null directions at the boundary of the feasible cone. However, if $\operatorname{rank}(\widetilde{H}_\Phi) < J$, any coefficient vector in the interior of $\mathbb{R}^J_+$ can be perturbed slightly along a null direction while remaining feasible. Full column rank is therefore the clean condition for interpretation at the selected basis resolution. Rank deficiency may arise from spatial source ambiguity, dependence among temporal modes, or their interaction.

5.2. Noise-Robust Identifiability

Let $\sigma_1(\widetilde{H}_\Phi) \geq \cdots \geq \sigma_J(\widetilde{H}_\Phi) \geq 0$ be the square roots of the eigenvalues of $\widetilde{H}_\Phi^\top \widetilde{H}_\Phi$, including zeros when $J > mT$ or the matrix is rank deficient. Even with full rank, small singular values amplify noise.

Proposition 5.3 (Noise-robust apportionment)**.** *Assume $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}$ and $\operatorname{rank}(\widetilde{H}_\Phi) = J$. Let $\widehat{\mathbf{c}}_{\mathrm{LS}}$ be the unconstrained least-squares estimate. Then*

$$\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 \leq \frac{\|\widetilde{E}\|_2}{\sigma_J(\widetilde{H}_\Phi)}. \tag{38}$$

If $\widehat{\mathbf{c}}$ is the nonnegative least-squares estimate and $\mathbf{c} \geq 0$, then

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{2\|\widetilde{E}\|_2}{\sigma_J(\widetilde{H}_\Phi)}. \tag{39}$$

Proof. For least squares, $\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c} = \widetilde{H}_\Phi^\dagger \widetilde{E}$, giving $\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 \leq \|\widetilde{H}_\Phi^\dagger\|_2 \|\widetilde{E}\|_2 = \|\widetilde{E}\|_2 / \sigma_J(\widetilde{H}_\Phi)$. For nonnegative least squares, the true $\mathbf{c}$ is feasible, so optimality gives $\|\widetilde{Y} - \widetilde{H}_\Phi \widehat{\mathbf{c}}\|_2 \leq \|\widetilde{E}\|_2$. Hence $\|\widetilde{H}_\Phi(\widehat{\mathbf{c}} - \mathbf{c})\|_2 \leq 2\|\widetilde{E}\|_2$. The lower singular-value bound completes the proof. □

Thus $\sigma_J(\widetilde{H}_\Phi)$ measures worst-case robustness: source apportionment can be exactly identifiable but practically unstable when this quantity is small.

5.3. Identifiability Diagnostics

For floating-point computation, the default numerical rank tolerance is

$$\begin{aligned} \tau_{\mathrm{num}} &= \max(mT, J)\, \epsilon_{\mathrm{mach}}\, \sigma_1, \\ r_{\mathrm{num}} &= \#\{i : \sigma_i > \tau_{\mathrm{num}}\}. \end{aligned} \tag{40}$$

The primary identifiability score includes rank-deficiency zeros:

$$\mathcal{I} = \sigma_J(\widetilde{H}_\Phi). \tag{41}$$

The condition number is

$$\kappa(\widetilde{H}_\Phi) = \begin{cases} \sigma_1/\sigma_J, & r_{\mathrm{num}} = J, \\ \infty, & r_{\mathrm{num}} < J. \end{cases} \tag{42}$$

The smallest numerically positive value may additionally be reported as $\sigma_{\min,+}$, but it is not the identifiability score and never turns a rank-deficient matrix into a stable one. For a separately chosen, noise-dependent threshold τ_σ, the effective rank is

$$r_{\mathrm{eff}}(\tau_\sigma) = \#\{i : \sigma_i(\widetilde{H}_\Phi) > \tau_\sigma\}. \tag{43}$$

The threshold τ_σ is recorded independently of τ_{num}. If $r_{\mathrm{eff}}(\tau_\sigma) < J$, the sensor network does not support reliable estimation of all J source–basis coefficients at that noise level.

For coefficient-specific diagnostics, let $\widetilde{\mathbf{h}}_j$, $j = (k, b)$, be a projected source–basis fingerprint. Its visibility and the weak set at threshold $\tau_v \geq 0$ are

$$v_j = \|\widetilde{\mathbf{h}}_j\|_2, \qquad \mathcal{W} = \{j : v_j \leq \tau_v\}. \tag{44}$$

A weak coefficient has little measurable effect after transport and background correction. For eligible indices $i, j \notin \mathcal{W}$, pairwise ambiguity is measured by projected fingerprint coherence,

$$\rho_{ij} = \frac{|\widetilde{\mathbf{h}}_i^\top \widetilde{\mathbf{h}}_j|}{\|\widetilde{\mathbf{h}}_i\|_2 \|\widetilde{\mathbf{h}}_j\|_2}. \tag{45}$$

Values near one indicate nearly proportional sensor-time signatures. The ambiguous-pair set is

$$\mathcal{A} = \{(i,j) : i,j \notin \mathcal{W},\ \rho_{ij} > \tau_\rho\}, \qquad \tau_\rho \in [0,1]. \tag{46}$$

Coherence is undefined when either fingerprint is weak. Such entries are reported as "N/A", excluded from coherence extrema, and never allowed to hide the corresponding weak-visibility flags.

To express the same scale-invariant geometry as a distance, we reactivate the ray distance for eligible, generally signed projected fingerprints:

$$d_{ij}^{\mathrm{ray}} = \min_{\alpha \in \mathbb{R}} \frac{\|\widetilde{\mathbf{h}}_i - \alpha \widetilde{\mathbf{h}}_j\|_2}{\|\widetilde{\mathbf{h}}_i\|_2}. \tag{47}$$

For nonweak fingerprints,

$$d_{ij}^{\mathrm{ray}} = \sqrt{1-\rho_{ij}^2}. \tag{48}$$

Numerical implementations evaluate $\sqrt{\max(0, 1-\rho_{ij}^2)}$. This is an unoriented linear-ray diagnostic, not a claim that projected fingerprints are nonnegative. High coherence and small ray distance encode the same pairwise geometry, so ray distance is a complementary report rather than an independent identifiability theorem. Like coherence, it is "N/A" for pairs involving $\mathcal{W}$ and excluded from distance extrema.

5.4. Background Absorption

Background correction can improve robustness by removing broad temporal variation, but it can also remove source-driven signal. Let $P_Q = QQ^\dagger$. For coefficient fingerprint $j = (k, b)$, define the background absorption fraction

$$a_j = \frac{\|P_Q \mathbf{h}_j^{\mathrm{lag}}\|_2}{\|\mathbf{h}_j^{\mathrm{lag}}\|_2}. \tag{49}$$

If $a_j \approx 1$, most of the raw source–basis fingerprint lies in the background space, and the projected visibility v_j will be small even if the raw fingerprint is large. This is why identifiability must be assessed after projection.

5.5. Response-Matrix Perturbations

The constructed response matrix can be wrong because of wind interpolation, transport approximation, dispersion parameters, or inventory error. Let $\widetilde{H}_\Phi^\star$ be the true projected response and $\widehat{\widetilde{H}}_\Phi$ the constructed one, with

$$\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \le \delta_H. \tag{50}$$

Then

$$\widetilde{Y} = \widehat{\widetilde{H}}_\Phi \mathbf{c} + \left[\widetilde{E} + (\widetilde{H}_\Phi^\star - \widehat{\widetilde{H}}_\Phi)\mathbf{c}\right]. \tag{51}$$

Response error therefore acts like additional observation error.

Table 1. Identifiability diagnostics reported with source activity estimates.

Diagnostic	Interpretation	Action
$r_{\mathrm{num}}(\widetilde{H}_\Phi)$	exact separability	require rank J
$\sigma_J(\widetilde{H}_\Phi)$	noise robustness	flag zero/small values
$\kappa(\widetilde{H}_\Phi)$	error amplification	set ∞ if deficient
r_{eff}	noise-level resolution	reduce grouping
v_j	coefficient visibility	flag weak coefficients
a_j	background absorption	warn after projection
ρ_{ij}	pairwise ambiguity	merge coherent groups
d_{ij}^{ray}	ray separation	report N/A if weak

Proposition 5.4 (Perturbation bound)**.** *Assume $\widehat{\widetilde{H}}_\Phi$ has full column rank and $\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \le \delta_H$. The unconstrained least-squares estimate using $\widehat{\widetilde{H}}_\Phi$ satisfies*

$$\|\hat{\mathbf{c}} - \mathbf{c}\|_2 \le \frac{\|\widetilde{E}\|_2 + \delta_H \|\mathbf{c}\|_2}{\sigma_J(\widehat{\widetilde{H}}_\Phi)}. \tag{52}$$

The effect of wind and transport uncertainty is therefore most damaging when the projected response matrix is already ill-conditioned.

5.6. Identifiable Source Resolution

When coefficient fingerprints are weak or highly coherent, separate source estimates can be misleading. Let $\mathcal{G} = \{G_1, \ldots, G_M\}$ be a partition of the original K source groups into merged groups, with merged inventory

$$S^{\mathcal{G}} = \left[\sum_{k \in G_1} \mathbf{s}_k \ \sum_{k \in G_2} \mathbf{s}_k \ \cdots \ \sum_{k \in G_M} \mathbf{s}_k\right]. \tag{53}$$

Let $\widetilde{H}_\Phi^{\mathcal{G}}$ be the corresponding projected source–basis response with MB columns. We say that $\mathcal{G}$ is identifiable at thresholds τ_σ and τ_ρ if

$$r_{\mathrm{eff}}(\widetilde{H}_\Phi^{\mathcal{G}}; \tau_\sigma) = MB, \qquad \max_{\substack{a \ne a' \\ b, b'}} \rho_{(a,b),(a',b')}^{\mathcal{G}} \le \tau_\rho. \tag{54}$$

The reporting rule is to use the finest partition satisfying these conditions, while flagging low-visibility coefficients and retaining the basis pair that triggers each source-level ambiguity. Matrix diagnostics remain coefficient level; source-level activity summaries are reconstructed from the fitted basis coefficients rather than from an artificial sum of fingerprint columns.

6. Identifiability-Aware Apportionment Method

We combine the estimator from Section 4 with the diagnostics from Section 5. The result is an identifiability-aware source apportionment procedure, abbreviated IASA, that estimates source contributions only at the resolution supported by the projected response matrix.

6.1. Inputs and Outputs

IASA takes sparse pollution observations Y, sensor locations X_{sens}, wind observations $\mathbf{w}^{\mathrm{obs}}_{1:T}$, source inventories $S \in \mathbb{R}^{q\times K}$, a background basis Q, a maximum lag L, a non-negative temporal basis $\Phi \in \mathbb{R}^{T\times B}_{+}$, and an open-boundary transport response G^{∂}. It also uses a numerical SVD tolerance τ_{num} and distinct thresholds τ_σ, τ_ρ, and τ_v for effective rank, pairwise ambiguity, and visibility.

The outputs are source–basis coefficient estimates, reconstructed source activities and contribution estimates at the identifiable grouping, fitted trajectories and residuals, uncertainty intervals, singular-value, coherence, and ray-distance diagnostics, low-visibility flags, and merge recommendations.

6.2. Procedure

The procedure is:

1. **Estimate wind.** Construct $\widehat{\mathbf{w}}_{1:T} = f_\phi(\mathbf{w}^{\mathrm{obs}}_{1:T}, X_{\mathrm{wind}}, X_{\mathrm{grid}})$.
2. **Build lagged responses.** For every source–basis pair, compute
$$\mathbf{h}^{\mathrm{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell) O G^{\partial}_{t,t-\ell}(\widehat{\mathbf{w}}_{t-\ell:t})\mathbf{s}_k.$$
3. **Stack and project.** Form H^{lag}_{Φ}, with source-major, basis-minor columns, then compute
$$\widetilde{Y} = P_Q^{\perp} Y, \qquad \widetilde{H}_\Phi = P_Q^{\perp} H^{\mathrm{lag}}_{\Phi}.$$
4. **Fit source activities.** Solve for coefficients and reconstruct activities:
$$\widehat{\mathbf{c}} = \arg\min_{\mathbf{c}\ge 0} \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda R(\mathbf{c}),$$
$$\widehat{\theta}_k(t) = \sum_b \widehat{c}_{kb}\phi_b(t).$$
5. **Audit identifiability.** Compute r_{num}, the singular values, $r_{\mathrm{eff}}(\tau_\sigma)$, coefficient visibility v_j, background absorption a_j, coherence ρ_{ij}, and ray distance d^{ray}_{ij}.
6. **Flag unsupported groups.** Mark weak coefficient fingerprints $\mathcal{W} = \{j : v_j \le \tau_v\}$. For eligible pairs only, set $\mathcal{A} = \{(i,j) : i,j \notin \mathcal{W}, \rho_{ij} > \tau_\rho\}$.
7. **Merge when necessary.** If $\mathcal{A} \neq \emptyset$, merge source groups connected by high-coherence edges, reconstruct the merged inventory $S^{\mathcal{G}}$, recompute $\widetilde{H}^{\mathcal{G}}_{\Phi}$, and refit contributions at the merged resolution.
8. **Report.** Return source estimates, uncertainty intervals, residuals, diagnostics, and the final source grouping.

This procedure separates two questions: which source activity trajectories best fit the projected sensor data, and which source–basis coefficients are supported by the geometry of $\widetilde{H}_\Phi$.

6.3. Uncertainty Reporting

Let
$$\widetilde{r} = \widetilde{Y} - \widetilde{H}_\Phi \widehat{\mathbf{c}} \tag{55}$$
be the projected residual. A residual variance estimate is
$$\widehat{\sigma}^2 = \frac{\|\widetilde{r}\|_2^2}{\max(mT - r_{\mathrm{eff}}(\tau_\sigma), 1)}. \tag{56}$$

For an unconstrained or active-set approximation, let $\mathcal{A}_c = \{j : \widehat{c}_j > \tau_c\}$ and let $\widetilde{H}_{\Phi,\mathcal{A}_c}$ contain the active columns. The covariance of the active estimates is approximated by
$$\mathrm{Cov}(\widehat{\mathbf{c}}_{\mathcal{A}_c}) = \widehat{\sigma}^2 (\widetilde{H}^{\top}_{\Phi,\mathcal{A}_c}\widetilde{H}_{\Phi,\mathcal{A}_c} + \lambda I)^{\dagger}. \tag{57}$$

When wind ensembles are available, IASA also constructs response matrices $\widetilde{H}^{(b)}_{\Phi}$, refits $\widehat{\mathbf{c}}^{(b)}$, and reports empirical quantiles across ensemble members. These intervals capture uncertainty induced by wind interpolation and transport-response construction.

6.4. Acceptance Criteria for Refinement

The constrained refinement in Section 4.4 is accepted only if it improves fit without degrading separability. Let $\widetilde{H}_{\Phi,0}$ be the projected response before refinement and $\widetilde{H}_{\Phi,\mathrm{ref}}$ after refinement. We require
$$\sigma_J(\widetilde{H}_{\Phi,\mathrm{ref}}) \ge (1-\eta_{\mathrm{id}})\sigma_J(\widetilde{H}_{\Phi,0}) \tag{58}$$
and
$$\max_{\substack{i\neq j\\ i,j\notin\mathcal{W}}} \rho^{\mathrm{ref}}_{ij} \le \tau^{\mathrm{ref}}_{\rho}. \tag{59}$$

These checks prevent end-to-end tuning from improving sensor fit by making source–basis fingerprints less distinguishable. If either response is numerically rank deficient, its σ_J is zero under the padded-spectrum convention of Section 5.3.

6.5. Reporting Protocol

For each reported source group G_a, IASA reports
$$\widehat{\boldsymbol{\theta}}_{G_a,1:T}, \quad \mathrm{CI}(\widehat{\theta}_{G_a}), \quad \{v_{kb} : k\in G_a\},$$
$$\max_{\substack{k\in G_a,\, k'\notin G_a\\ (k,b),(k',b')\notin\mathcal{W}}} \rho_{(k,b),(k',b')}. \tag{60}$$

A source contribution is marked reliable only when its constituent reported coefficients are above the visibility threshold, its uncertainty interval is sufficiently narrow, and its eligible coherence with other reported groups is below the ambiguity threshold. Reports retain the basis names of every weak coefficient and the coefficient pair attaining each cross-source coherence maximum. Coherence and ray distance involving weak fingerprints are displayed as “N/A” and excluded from extrema; the weak flags remain visible. Otherwise IASA labels the source as weakly visible, ambiguous with another source group, or merged into a coarser identifiable group.

7. Empirical Validation

The empirical study evaluates whether source attribution error is predicted by the identifiability structure of $\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}}$. The experiments use a controlled city-scale advection–diffusion simulator so that ground-truth source activities, wind regimes, noise levels, background components, and sensor layouts can be varied independently.

7.1. Pollution Simulator and Sensor Layout

We simulate pollutant transport on a normalized $[0,1]\times[0,1]$ domain corresponding to the New Delhi region using

$$\partial_t u + V_x(t)\partial_x u + V_y(t)\partial_y u = k\Delta u + S(x,y,t). \quad (61)$$

The state is discretized on a 40×40 grid. Advection is approximated with first-order upwind differences, diffusion with a five-point Laplacian stencil, and time integration with a two-stage Heun update. The base wind is $(V_x, V_y) = (1.12, 0.984)$, the diffusivity is $k = 3 \times 10^{-4}$, and the wind includes diurnal modulation and AR(1) variability. The rollout horizon is $T = 20$ with $N_t = 10{,}000$ integration steps, giving $\Delta t \approx 2 \times 10^{-3}$.

Sensor locations are obtained by mapping regulatory monitoring sites from the New Delhi region to the simulation grid. The raw location file contains 33 sites, but one pair collapses to the same grid cell, yielding $m = 32$ unique sensor locations. With $q = 40\times 40 = 1600$ source cells, the sensor density is $m/q = 0.02$. Initial conditions are constructed from government monitoring data using Universal Kriging rather than learned jointly, so that the experiments focus on source attribution rather than initial-condition recovery (CPCB, 2023; 2025).

7.2. Inventories and Ground Truth

The source-inventory matrix $S = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_K] \in \mathbb{R}^{q\times K}$ is built from proxy maps, including brick-kiln and industrial emissions (Guttikunda & Calori, 2013), population density (Columbia University, 2018), and traffic intensity derived from map data (Google, 2024). The maps are cropped from 80×80 to the 40×40 simulation window and normalized. At minimum, traffic, industrial or brick-kiln activity, and population-related emissions are treated as separate source groups. To create controlled ambiguity, inventories can also be split into spatial subgroups such as north/south, near/far, or upwind/downwind regions.

Ground-truth observations are generated by choosing nonnegative temporal-basis coefficients $C^\star \in \mathbb{R}_+^{K\times B}$, reconstructing $\theta_k^\star(t) = \sum_b c_{kb}^\star \phi_b(t)$, and simulating

$$S(x,y,t) = \sum_{k=1}^{K} \theta_k^\star(t) s_k(x,y). \quad (62)$$

We evaluate sparse, dense, balanced, and near-collinear coefficient mixtures, including the constant-activity case $B = 1$, to test both identifiable and ambiguous regimes.

7.3. Noise and Background Settings

Clean sensor trajectories $Y_{\mathrm{clean}} \in \mathbb{R}^{m\times T}$ are perturbed by Gaussian observation noise,

$$Y_{\mathrm{noisy}} = Y_{\mathrm{clean}} + \epsilon, \qquad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \quad (63)$$

with σ set to 1%, 5%, 10%, or 20% of the maximum clean sensor signal. Background components are generated from low-dimensional temporal and sensor bases, including smooth time-of-day variation, day-level intercepts, regional trends, and sensor offsets. This allows comparison among no correction, moderate correction, and overly flexible correction.

7.4. H1: Conditioning Predicts Attribution Error

The first validation tests the central prediction that coefficient and reconstructed-activity errors are controlled by the conditioning of $\widetilde{H}_\Phi$. For each trial, we construct H_Φ^{lag}, project by $P_Q^\perp$, fit $\widehat{\mathbf{c}}$, reconstruct $\widehat{\theta}_k(t)$, and compute

$$\|\widehat{\mathbf{c}} - \mathbf{c}^\star\|_2, \qquad \|\widehat{\Theta} - \Theta^\star\|_F. \quad (64)$$

We compare these errors with $\sigma_J(\widetilde{H}_\Phi)$, $\kappa(\widetilde{H}_\Phi)$, $r_{\mathrm{eff}}(\tau_\sigma)$, and coefficient visibility v_j. The expected pattern is increasing attribution error as σ_J decreases, κ increases, or effective rank falls below J. Activity errors should be largest for sources containing weak source–basis fingerprints.

7.5. H2: Coherent Sources Should Be Merged

The second validation tests whether pairwise coherence identifies source groups that should not be interpreted separately. We construct source pairs with increasing ρ_{ij} by placing groups along similar transport paths, splitting a single inventory into nearby subregions, or creating near/far pairs whose fingerprints differ mainly by scale. For each pair, we compare separate estimation error with merged-group activity-trajectory error,

$$\left\| (\widehat{\boldsymbol{\theta}}_{i,1:T} + \widehat{\boldsymbol{\theta}}_{j,1:T}) - (\boldsymbol{\theta}^{\star}_{i,1:T} + \boldsymbol{\theta}^{\star}_{j,1:T}) \right\|_2 . \quad (65)$$

High-coherence pairs should exhibit unstable separate estimates but more stable combined estimates, validating the merge rule.

7.6. H3: Background Correction Can Help or Hurt

The third validation studies how the background basis changes identifiability. For each coefficient fingerprint $j = (k, b)$, we compute

$$a_j = \frac{\|P_Q \mathbf{h}_j^{\mathrm{lag}}\|_2}{\|\mathbf{h}_j^{\mathrm{lag}}\|_2}. \quad (66)$$

Moderate background correction should improve robustness when it removes broad temporal confounding. Overly flexible correction should reduce projected visibility and increase attribution error for coefficients with large absorption fractions.

7.7. H4: Wind Diversity and Sensor Geometry Change Resolution

The fourth validation varies the wind regime and sensor layout. Wind settings include constant wind, diurnally varying wind, AR(1)-perturbed wind, and multi-directional wind episodes. Sensor settings include the regulatory layout, random layouts, downwind-focused layouts, and layouts selected to increase $\sigma_J(\widetilde{H}_\Phi)$ or reduce maximum eligible coherence. For each setting, we report singular values, effective rank, visibility, coherence, ray distance, and source activity error. Diverse wind and strategically placed sensors should improve the identifiable temporal source resolution more than spatial coverage alone.

7.8. H5: Response-Matrix Error Amplifies Attribution Error

The final validation perturbs the constructed response matrix through wind and transport errors. We add controlled errors to wind speed and direction, construct $\widehat{\widetilde{H}}_\Phi$, refit source activities, and compare attribution error with

$$\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}^{\star}_\Phi\|_2. \quad (67)$$

When wind ensembles are available, we construct $\widetilde{H}^{(b)}_\Phi$, refit $\widehat{\mathbf{c}}^{(b)}$, reconstruct activities, and report empirical uncertainty intervals. The perturbation bound predicts that response-matrix error is most damaging when $\widehat{\widetilde{H}}_\Phi$ is ill-conditioned.

7.9. Metrics

Across experiments, we report both attribution accuracy and identifiability structure. Accuracy metrics include absolute and relative coefficient error, source activity-trajectory error, per-source error, merged-group error, and fitted sensor-trajectory error $\|\widetilde{Y} - \widetilde{H}_\Phi \widehat{\mathbf{c}}\|_2$. Identifiability metrics include numerical rank, σ_J, condition number, effective rank, coefficient visibility, coherence, ray distance, background absorption, and merge recommendations. Coherence and ray distance are N/A for pairs involving $v_j \le \tau_v$; those entries are excluded from extrema while weak coefficients remain explicitly flagged. Cross-source summaries retain the eligible coefficient pair attaining maximum coherence and minimum ray distance.

The empirical claim is that source apportionment error is systematic rather than arbitrary: well-conditioned and visible fingerprints should be accurately apportioned, weakly visible coefficients should produce unstable activity components, high-coherence sources should be reported after merging, and wind or background uncertainty should reduce the identifiable temporal source resolution.

8. Discussion and Conclusion

This work frames sparse-sensor source apportionment as an identifiable-resolution problem. Known or proxy source inventories restrict the source space and make attribution meaningful, but they do not guarantee that source groups are separable from sparse observations. The separability question is governed by how inventory groups appear at the sensor network after wind-conditioned transport, finite lag, background correction, and noise.

The central object is the projected lagged source–basis response matrix

$$\widetilde{H}_\Phi = P_Q^{\perp} H_\Phi^{\mathrm{lag}} \in \mathbb{R}^{mT \times J}, \qquad J = KB. \quad (68)$$

Exact identifiability at the selected temporal-basis resolution requires $\mathrm{rank}(\widetilde{H}_\Phi) = J$, while robust attribution depends on the singular values (including rank-deficiency zeros), effective rank, coefficient visibility, background absorption, pairwise coherence, ray distance, and perturbation sensitivity of $\widetilde{H}_\Phi$. The constant-activity source model is the special case $B = 1$. A model can fit sensor trajectories while still producing unstable or non-unique coefficient and activity estimates, so prediction accuracy alone is not evidence of identifiability at the claimed temporal resolution.

The practical implication is that source apportionment should report the finest source grouping that the sensing system supports. If two source groups have highly coherent projected fingerprints, the data may support their combined contribution but not their separate activities. Reporting them separately can create false precision. The proposed IASA protocol therefore estimates nonnegative source–basis coefficients, reconstructs source activity trajectories, audits the projected response matrix, flags weak coefficients and ambiguous source pairs, and merges source groups when the fine-grained inventory resolution is not identifiable. Each source-level warning retains the temporal basis components that triggered it.

This perspective also changes how sensing systems and preprocessing choices should be designed. Adding sensors is useful when it improves visibility, conditioning, coherence, or ray separation of coefficient fingerprints, not merely when it increases spatial coverage. Wind diversity can expose source groups from multiple transport angles, while a single dominant wind regime can leave sources confounded. Background correction can remove broad temporal confounding, but a basis that is too flexible can absorb source-driven variation and reduce identifiability. Wind interpolation and transport-model errors act as additional observation error whose impact is amplified by ill-conditioning.

The framework has several limitations. The response operator is an approximation to atmospheric transport and can be extended with richer meteorology, vertical mixing, deposition, chemistry, and boundary inflow. The theory is linear in source–basis coefficients conditional on fixed wind, transport, temporal basis, inventory, and background components; fully joint estimation introduces additional nonlinearities. The selected temporal basis also limits the activity patterns that can be recovered: finer temporal resolution is defensible only when the corresponding response columns remain identifiable. Finally, the framework assumes that relevant candidate source inventories are available. Missing source categories may be absorbed into available inventories or background terms, which should be treated as a model-misspecification risk rather than as identifiable attribution.

Despite these limitations, the identifiability view provides a concrete basis for more reliable policy-facing apportionment. It separates fitting the observed sensor signal from determining which source claims are supported by the sensing system. By using rank, singular values, visibility, background absorption, coherence, ray distance, and response perturbation sensitivity, source apportionment can report not only estimated contributions, but also the spatial and temporal resolution at which those contributions can be trusted.

Acknowledgements

Ankit Bhardwaj and Lakshminarayanan Subramanian were supported by the NSF Grant (Award Number 2335773) titled “EAGER: Scalable Climate Modeling using Message-Passing Recurrent Neural Networks”. Lakshminarayanan Subramanian was also funded in part by the NSF Grant (award number OAC-2004572) titled “A Data-informed Framework for the Representation of Sub-grid Scale Gravity Waves to Improve Climate Prediction”.

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A. Temporal-Basis Notation and Diagnostic Conventions

Table 2. Indices and objects in the temporal-basis apportionment model.

Symbol	Shape or range	Meaning
k	$1, \dots, K$	source-group index
b	$1, \dots, B$	temporal-basis index
$j = (k, b)$	$1, \dots, J$	source-major, basis-minor coefficient index
Φ	$T \times B$	nonnegative source-activity basis
C	$K \times B$	nonnegative source–basis coefficients
$\mathbf{c}$	$J = KB$	source-major vectorization of C
H_{Φ}^{lag}	$mT \times J$	unprojected coefficient fingerprints
$\widetilde{H}_{\Phi}$	$mT \times J$	background-projected fingerprints
$\widetilde{\mathbf{h}}_{kb}$	mT	fingerprint for coefficient c_{kb}

The constant-activity model uses $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$. In the general model, diagnostics are computed for the J coefficient fingerprints. Source-level activity trajectories are reconstructed as

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb} \phi_b(t);$$

fingerprint columns are not summed to create an artificial source-level diagnostic vector.

For a source-level ambiguity summary, IASA reports the largest eligible coherence between coefficients belonging to different sources and retains the source–basis pair that attains it. Coherence and ray distance are undefined for pairs involving a coefficient with $v_j \leq \tau_v$; tables display these entries as N/A, exclude them from pairwise extrema, and report the weak coefficient separately.